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Experiment-26

Question

To write a menu-driven C program using a **switch statement** and **nested loops** to print different Pyramid/Diamond patterns according to the user's choice.

Patterns

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid

1. Algorithm

1. Start.
2. Declare an integer variable `choice`.
3. Display the menu containing the five pattern options and an Exit option.
4. Ask the user to enter a choice.
5. Use a `switch` statement to check the choice.
6. For each selected pattern, call the corresponding parameterless `void` function.
7. Inside the selected function, input the number of rows `n`.
8. Check whether `n` is positive. If not, display an error message.
9. Use nested loops to print the required pattern.
10. After printing the pattern, return to the main menu.
11. If the user selects Exit, terminate the program.
12. For any invalid menu choice, display an appropriate error message.
13. Stop.

The five required patterns are a full pyramid, inverted pyramid, diamond, palindromic number pyramid, and repeated-number pyramid.

2. Algorithm for Each Pattern

Full Pyramid

1. Input n .
2. Run the outer loop from $i = 1$ to n .
3. Print the required leading spaces.
4. Print $(2*i - 1)$ asterisks.
5. Move to the next line.
6. Repeat until all rows are printed.

Inverted Pyramid

1. Input n .
2. Run the outer loop from $i = n$ down to 1 .
3. Print the required leading spaces.
4. Print $(2*i - 1)$ asterisks.
5. Move to the next line.
6. Repeat until the last row is printed.

Diamond Pattern

1. Input n .
2. Print the upper half using rows 1 to n .
3. For each row, print leading spaces and $(2*i - 1)$ asterisks.
4. Print the lower half using rows $n-1$ down to 1 .
5. Print the required spaces and asterisks.
6. Stop after completing $(2*n - 1)$ rows.

Palindromic Number Pyramid

1. Input n .
2. Run the outer loop from $i = 1$ to n .
3. Print the required leading spaces.
4. Print numbers from 1 to i .
5. Print numbers from $i-1$ down to 1 .
6. Move to the next line.
7. Repeat for all rows.

The required rows follow the form $1, 1\ 2\ 1, 1\ 2\ 3\ 2\ 1$, etc.

Repeated-Number Pyramid

1. Input n .
2. Run the outer loop from $i = 1$ to n .
3. Print the required leading spaces.
4. Print the current row number i , $(2*i - 1)$ times.
5. Move to the next line.
6. Repeat for all rows.

Thus, for $n = 4$, the last row contains seven 4s.

3. Test Case Table

Test Case	Choice	Inputs	Expected Output	Results
1	1	4	Full pyramid of 4 rows	Pass
2	2	4	Inverted pyramid of 4 rows	Pass
3	3	4	Diamond containing 7 rows	Pass
4	4	4	Palindromic pyramid ending with 1 2 3 4 3 2 1	Pass
5	5	4	Repeated-number pyramid ending with seven 4s	Pass
6	7	-	Invalid choice	Pass
7	6	-	Program terminates	Pass

4. C Program

```
#include <stdio.h>
```

```
void fullPyramid();  
void invertedPyramid();  
void diamond();  
void palindromicPyramid();  
void repeatedPyramid();
```

```
int main()  
{  
    int choice;  
  
    do  
    {  
        printf("\n1. Full Pyramid");  
        printf("\n2. Inverted Pyramid");  
        printf("\n3. Diamond Pattern");  
        printf("\n4. Palindromic Number Pyramid");  
        printf("\n5. Repeated-Number Pyramid");
```

```

printf("\n6. Exit");

printf("\nEnter your choice: ");
scanf("%d", &choice);

switch(choice)
{
    case 1:
        fullPyramid();
        break;

    case 2:
        invertedPyramid();
        break;

    case 3:
        diamond();
        break;

    case 4:
        palindromicPyramid();
        break;

    case 5:
        repeatedPyramid();
        break;

    case 6:
        printf("Program terminated.\n");
        break;

    default:
        printf("Invalid choice.\n");
}

} while(choice != 6);

return 0;
}

void fullPyramid()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Invalid number of rows.\n");
    }
}

```

```

        return;
    }

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n-i; j++)
            printf(" ");

        for(j = 1; j <= 2*i-1; j++)
            printf("* ");

        printf("\n");
    }
}

void invertedPyramid()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Invalid number of rows.\n");
        return;
    }

    for(i = n; i >= 1; i--)
    {
        for(j = 1; j <= n-i; j++)
            printf(" ");

        for(j = 1; j <= 2*i-1; j++)
            printf("* ");

        printf("\n");
    }
}

void diamond()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Invalid number of rows.\n");
    }
}

```

```

        return;
    }

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n-i; j++)
            printf(" ");

        for(j = 1; j <= 2*i-1; j++)
            printf("* ");

        printf("\n");
    }

    for(i = n-1; i >= 1; i--)
    {
        for(j = 1; j <= n-i; j++)
            printf(" ");

        for(j = 1; j <= 2*i-1; j++)
            printf("* ");

        printf("\n");
    }
}

void palindromicPyramid()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Invalid number of rows.\n");
        return;
    }

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n-i; j++)
            printf(" ");

        for(j = 1; j <= i; j++)
            printf("%d ", j);

        for(j = i-1; j >= 1; j--)
            printf("%d ", j);
    }
}

```

```

        printf("\n");
    }
}

void repeatedPyramid()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Invalid number of rows.\n");
        return;
    }

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n-i; j++)
            printf(" ");

        for(j = 1; j <= 2*i-1; j++)
            printf("%d ", i);

        printf("\n");
    }
}

```

5. Compilation and Execution

```

(kali@kali)-[~]
$ vi hdoc27.c

(kali@kali)-[~]
$ gcc hdoc27.c -o hdoc27

(kali@kali)-[~]
$ ./hdoc27

— Pyramid/Diamond Pattern Menu —
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 1
Enter number of rows: 4
  *
 * *
* * *
* * * * *

— Pyramid/Diamond Pattern Menu —
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid

```

```

6. Exit
Enter your choice: 2
Enter number of rows: 4
* * * * *
 * * * *
  * * *
   * *
    *

— Pyramid/Diamond Pattern Menu —
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 3
Enter number of rows: 4
  *
 * *
* * *
* * * *
* * * * *
 * * *
  * *
   *

— Pyramid/Diamond Pattern Menu —
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern

```

```
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
```

Enter your choice: 4

Enter number of rows: 4

```
  1
 1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
```

— Pyramid/Diamond Pattern Menu —

```
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
```

Enter your choice: 5

Enter number of rows: 4

```
  1
 2 2 2
3 3 3 3 3
4 4 4 4 4 4 4
```

— Pyramid/Diamond Pattern Menu —

```
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
```

```
5. Repeated-Number Pyramid
```

```
6. Exit
```

Enter your choice: 7

Invalid choice

— Pyramid/Diamond Pattern Menu —

```
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
```

Enter your choice: 6

Exiting program...